MTH166

Lecture-22

Classification of PDE

Unit 4: **Partial Differential Equations**

(**Book:** Advanced Engineering Mathematics by R.K.Jain and S.R.K Iyengar,**Chapter-9**)

Topic:

Partial Differential Equations (PDE)

Learning Outcomes:

1. Classification of PDE: Hyperbolic, Parabolic and Elliptic

Classification/Nature of Partial Differential Equations:

Let us consider a general second order homogeneous PDE:

$$A\frac{\partial^2 u}{\partial x^2} + B\,\frac{\partial^2 u}{\partial x \partial y} + \mathrm{C}\,\frac{\partial^2 u}{\partial y^2} + D\,\frac{\partial u}{\partial x} + E\,\frac{\partial u}{\partial y} + Fu = 0 \quad \text{where} \quad u = f(x, y)$$

or

$$Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = 0 \qquad (1)$$

or $Ar + Bs + Ct + f(x, y, u, p, q) = 0$ (1)

Equation (1) will be:

1. Hyperbolic iff $(B^2 - 4AC) > 0$

2. Parabolic iff $(B^2 - 4AC) = 0$

3. Elliptic iff $(B^2 - 4AC) < 0$

Note: Coefficients of second order partial derivatives only decide nature of PDE.

Classify the following PDE as Hyperbolic, Parabolic or Elliptic:

Problem 1. $\frac{\partial^2 u}{\partial x \partial y} = 3\frac{\partial u}{\partial y}$

Solution. Given equ. is: $u_{xy} - 3u_y = 0$ (1)

Comparing with: $Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = 0$

We have $A = 0, B = 1, C = 0$

Here $(B^2 - 4AC) = (1)^2 - 4(0)(0) = 1 > 0$

So, equation (1) is Hyperbolic.

Problem 2. $\frac{\partial^2 u}{\partial x^2} + 2\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$

Solution. Given equ. is: $u_{xx} + 2u_{xy} + u_{yy} = 0$ (1)

Comparing with: $Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = 0$

We have $A = 1, B = 2, C = 1$

Here $(B^2 - 4AC) = (2)^2 - 4(1)(1) = 4 - 4 = 0$

So, equation (1) is Parabolic.

Problem 3. $\frac{\partial^2 u}{\partial x^2} + 3\frac{\partial^2 u}{\partial y^2} = \frac{\partial u}{\partial x}$

Solution. Given equ. is: $u_{xx} + 3u_{yy} - u_x = 0$ (1)

Comparing with: $Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = 0$

We have $A = 1, B = 0, C = 3$

Here $(B^2 - 4AC) = (0)^2 - 4(1)(3) = 0 - 12 < 0$

So, equation (1) is Elliptic.

Polling Quiz

Let us consider a general second order homogeneous PDE:

$$Ar + Bs + Ct + f(x, y, u, p, q) = 0 \qquad (1)$$

Equation (1) will be Hyperbolic iff :

(A) $(B^2 - 4AC) > 0$

(B) $(B^2 - 4AC) = 0$

(C) $(B^2 - 4AC) < 0$

Problem 4. $y\frac{\partial^2 u}{\partial x^2} + 2x\frac{\partial^2 u}{\partial x \partial y} + y\frac{\partial^2 u}{\partial y^2} = 0$

Solution. Given equ. is: $yu_{xx} + 2xyu_{xy} + yu_{yy} = 0$ (1)

Comparing with: $Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = 0$

We have $A = y, B = 2x, C = y$

Here $(B^2 - 4AC) = (2x)^2 - 4(y)(y) = 4(x^2 - y^2)$

So, equation (1) is:

Hyperbolic iff $(x^2 - y^2) > 0$

Parabolic iff $(x^2 - y^2) = 0$

Elliptic iff $(x^2 - y^2) < 0$

Problem 5. $\boldsymbol{\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0}$ **[2D-Laplace Equation]**

Solution. Given equ. is: $u_{xx} + u_{yy} = 0$ (1)

Comparing with: $Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = 0$

We have $A = 1, B = 0, C = 1$

Here $(B^2 - 4AC) = (0)^2 - 4(1)(1) = -4 < 0$

So, **Laplace** equation (1) is **Elliptic** in nature.

Problem 6. $\boldsymbol{\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}}$ **[1D-Wave Equation]; $\boldsymbol{c^2}$ is Diffusivity constant.**

Solution. Given equ. is: $c^2\, u_{xx} - u_{tt} = 0 \qquad (1)$

Comparing with: $Au_{xx} + Bu_{xt} + Cu_{tt} + Du_x + Eu_t + Fu = 0$

We have $A = c^2, B = 0, C = -1$

Here $(B^2 - 4AC) = (0)^2 - 4(c^2)(-1) = 4c^2 > 0$

So, **Wave** equation (1) is **Hyperbolic** in nature.

Problem 7. $\boldsymbol{\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}}$ **[1D-Heat Equation]; $\boldsymbol{c^2}$ is Diffusivity constant.**

Solution. Given equ. is: $c^2\, u_{xx} - u_t = 0$ (1)

Comparing with: $Au_{xx} + Bu_{xt} + Cu_{tt} + Du_x + Eu_t + Fu = 0$

We have $A = c^2, B = 0, C = 0$

Here $(B^2 - 4AC) = (0)^2 - 4(c^2)(0) = 0$

So, **Heat** equation (1) is **Parabolic** in nature.

<u>**Polling Quiz**</u>

The nature of PDE: $(1-\mathrm{y})\frac{\partial^2 u}{\partial x^2}+2x\frac{\partial^2 u}{\partial x\partial y}+(1+\mathrm{y})\frac{\partial^2 u}{\partial y^2}=0$ is:

(A) Hyperbolic

(B) Parabolic

(C) Elliptic

(D) All of above is possible

Problem 8. $(\mathbf{1-y})\frac{\partial^2 u}{\partial x^2} + 2x\frac{\partial^2 u}{\partial x \partial y} + (\mathbf{1+y})\frac{\partial^2 u}{\partial y^2} = \mathbf{0}$

Solution. Given equ. is: $(1-y)u_{xx}+2xu_{xy} + (1+y)u_{yy} = 0$ (1)

Comparing with: $Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = 0$

We have $A = (1-y), B = 2x, C = (1+y)$

Here $(B^2 - 4AC) = (2x)^2 - 4(1-y)(1+y) = 4(x^2 + y^2 - 1)$

So, equation (1) is:

Hyperbolic iff $(x^2 + y^2 - 1) > 0$

Parabolic iff $(x^2 + y^2 - 1) = 0$

Elliptic iff $(x^2 + y^2 - 1) < 0$

Polling Quiz

The nature of one dimensional Heat Equation is:

(A) Hyperbolic

(B) Parabolic

(C) Elliptic

Polling Quiz

The nature of one dimensional Wave Equation is:

(A) Hyperbolic

(B) Parabolic

(C) Elliptic

<u>**Polling Quiz**</u>

The nature of two dimensional Laplace Equation is:

(A) Hyperbolic

(B) Parabolic

(C) Elliptic